

Rules of Engagement (RoE) Document

Course: WIC2007 Cyber Security — Alternative Assessment
Session: Academic Session 2025/2026 Semester 2
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Date: 16 May 2026

1. Introduction

This Rules of Engagement document defines the scope, boundaries, and operational constraints for the AI agent conducting penetration testing as part of the WIC2007 Alternative Assessment.

2. Test Scope

Field	Details
Target Application	Damn Vulnerable Web Application (DVWA)
Target IP Address	172.23.0.6
Permitted Ports	Port 80 (HTTP) only
Application URL	http://172.23.0.6:80
Environment	Isolated Docker network (pentagi-network)
Testing Period	May — June 2026

3. Permitted Activities

- Port scanning and service enumeration
- Web application vulnerability scanning
- SQL Injection testing
- Cross-Site Scripting (XSS) testing
- Authentication bypass testing
- Directory enumeration
- HTTP header analysis

4. Prohibited Techniques

The following techniques are strictly **PROHIBITED** for the AI agent:

4.1 Denial of Service (DoS) Attacks

The agent must NOT perform any flooding, resource exhaustion, or availability-disrupting attacks against the target or any other system.

4.2 Attacks Against Out-of-Scope Targets

The agent must ONLY target 172.23.0.6. Any scanning or exploitation attempt against other IP addresses, including the host machine or external internet targets, is strictly prohibited.

4.3 Destructive Exploitation

The agent must NOT delete, corrupt, or permanently modify any data or system files on the target beyond what is necessary to demonstrate vulnerability.

4.4 Privilege Escalation Beyond Web Application

The agent must NOT attempt OS-level privilege escalation, container breakout, or lateral movement beyond the designated DVWA container.

4.5 Social Engineering

The agent must NOT perform phishing, pretexting, or any human-targeted manipulation techniques.

5. Budget Constraints

Field	Details
Maximum Token Budget	\$0.50 USD per session
LLM Model	qwen/qwen2.5-72b-instruct
Input Cost	\$0.04 per 1M tokens
Output Cost	\$0.1 per 1M tokens
Kill-switch	Agent automatically halts when cumulative cost reaches \$0.50

6. Escalation Procedure

If the AI agent attempts any prohibited action, the following steps must be taken immediately:

Step 1 — IMMEDIATE HALT

Terminate the agent session immediately via the PentAGI interface or budget kill-switch.

Step 2 — LOG THE INCIDENT

Record the prohibited action, timestamp, and tool used in `agent_execution_log.json`.

Step 3 — REVIEW & ANALYSE

Review the Neo4j knowledge graph to identify what triggered the prohibited behaviour.

Step 4 — RECONFIGURE & RESTART

Adjust the agent system prompt to add explicit restrictions before restarting the session.

7. Authorisation

I confirm that this penetration test is conducted solely within the designated Docker environment for academic purposes under WIC2007 Cyber Security assessment.

Field	Details
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Date	16 May 2026
Authorised by	WIC2007 Course Lecturer, Universiti Malaya